

ANAKTUVUK PASS, AK, USA

WMO: 701625

Lat: 68.135N

Lon: 151.742W

Elev: 640

StdP: 93.87

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26542

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.7	(c) -36.2	(d) -34.5	(e) 0.2	(f) -28.3	(g) -34.0	(h) 0.2	(i) -28.3	(j) 12.0	(k) -18.7	(l) 11.1	(m) -20.8	(n) 3.3	(o) 30	(p) 0.912

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	21.6	12.9	19.3	12.0	17.7	11.1	13.7	19.7	12.8	18.2	11.8	16.6	4.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	8.9	15.5	9.9	8.2	14.6	8.8	7.6	14.2	40.2	19.8	37.7	18.0	35.4	17.0	17.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.9	(b) 8.8	(c) 8.1	(e) -40.3	(f) 23.9	(g) 2.4	(h) 1.2	(i) -42.1	(j) 24.8	(k) -43.5	(l) 25.5	(m) -44.8	(n) 26.2	(o) -46.6	(p) 27.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.6	-22.4	-20.4	-19.0	-11.6	0.5	9.8	11.8	7.4	1.2	-10.2	-17.8	-21.3		
	DBStd	14.14	8.63	8.63	7.69	7.46	6.64	4.14	3.29	3.89	5.16	6.58	6.86	7.79		
	HDD10.0	6558	1003	851	898	647	296	52	22	98	265	625	834	969		
	HDD18.3	9458	1262	1084	1156	897	553	257	202	339	514	883	1084	1227		
	CDD10.0	142	0	0	0	0	1	45	78	17	1	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	11	0	0	0	0	0	4	7	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.9	4.3	4.3	4.2	3.8	3.8	3.8	3.6	3.7	3.4	3.6	4.0	4.2		
(15) Precipitation	PrecAvg	302	19	18	12	10	15	30	45	50	35	28	21	21		
	PrecMax	423	51	49	30	46	39	73	88	127	105	58	57	75		
	PrecMin	189	3	1	2	1	4	9	21	17	7	6	0	3		
	PrecStd	55	12	13	8	10	7	13	16	22	22	11	15	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.2	-0.9	0.3	6.1	17.8	23.7	24.2	21.9	14.5	7.3	-1.0	-2.3		
		MCWB	-3.8	-2.5	-1.5	2.5	9.4	12.8	14.4	13.5	8.0	3.4	-2.3	-3.4		
	2%	DB	-4.3	-2.8	-2.6	2.9	15.0	21.4	22.1	17.8	12.4	2.7	-2.8	-5.0		
		MCWB	-5.4	-4.2	-4.0	0.1	8.0	12.0	13.6	11.4	7.2	0.9	-4.1	-6.0		
	5%	DB	-7.2	-6.1	-5.7	1.4	12.5	19.0	20.2	16.1	10.9	1.1	-5.1	-7.5		
		MCWB	-8.1	-7.3	-6.9	-0.8	6.7	11.1	12.8	10.4	6.7	-0.5	-6.2	-8.3		
	10%	DB	-9.0	-8.2	-8.1	-0.8	10.2	17.5	18.0	13.8	8.6	-1.1	-7.6	-10.0		
		MCWB	-9.8	-9.0	-9.0	-2.6	5.6	10.4	12.0	9.3	5.7	-2.2	-8.5	-10.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.5	-2.3	-1.3	2.6	10.2	14.1	15.5	14.0	9.1	4.0	-2.1	-3.2		
		MCDB	-2.3	-0.8	0.6	5.9	17.0	21.2	21.4	19.8	13.7	6.1	-1.4	-2.3		
	2%	WB	-5.2	-4.0	-3.7	0.6	8.5	12.9	14.1	12.4	7.8	1.1	-4.1	-5.5		
		MCDB	-4.1	-2.7	-2.4	2.8	14.5	19.3	20.1	17.0	11.5	2.7	-2.8	-4.7		
	5%	WB	-7.4	-6.5	-6.4	-0.7	7.1	11.9	13.3	10.9	6.8	-0.6	-6.1	-8.0		
		MCDB	-6.6	-5.3	-5.3	1.4	11.9	17.4	18.9	14.9	9.9	0.8	-4.9	-7.2		
	10%	WB	-9.7	-8.7	-9.0	-2.3	5.6	10.9	12.5	9.8	5.7	-2.2	-8.6	-10.2		
		MCDB	-9.0	-7.8	-8.0	-0.7	9.9	16.6	17.5	13.5	8.4	-1.2	-7.8	-9.4		
(35) Mean Daily Temperature Range	MDBR		6.9	7.2	7.9	9.3	9.8	10.1	9.3	8.3	7.2	5.9	6.1	6.6		
	5% DB	MCDBR	8.8	8.3	9.1	9.3	13.7	13.8	13.3	12.2	11.0	6.2	6.5	9.0		
		MCWBR	8.1	7.6	8.1	7.4	8.1	6.6	6.4	6.9	6.7	4.8	6.0	8.3		
	5% WB	MCDBR	9.1	8.4	8.8	9.1	13.0	12.0	11.6	10.0	9.5	5.7	6.4	8.6		
		MCWBR	8.5	7.7	7.9	7.2	8.0	6.1	5.9	5.9	6.0	4.5	6.0	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.157	0.219	0.244	0.304	0.342	0.344	0.350	0.326	0.294	0.217	0.156	N/A		
	taud		1.909	2.108	2.239	2.117	2.163	2.382	2.402	2.473	2.449	2.212	1.917	N/A		
	Ebn at Noon		138	603	815	837	841	856	837	822	757	603	116	0		
	Edh at Noon		8	45	73	112	121	101	96	80	62	38	7	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.53	2.28	4.53	5.66	5.56	4.68	3.39	2.01	0.75	0.11	0.00		
	RadStd		0.00	0.05	0.14	0.22	0.45	0.51	0.44	0.36	0.23	0.09	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+0.84	N/A	N/A	N/A	N/A	N/A	N/A	-274	N/A	N/A

Nomenclature: See separate page